

Matter as incomplete crystallization: Quark charges, color confinement, and the proton mass from a single extra node in the vacuum lattice

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ABSTRACT

We propose that baryonic matter is spacetime that failed to fully crystallize. In the Selection-Stitch Model, the early universe undergoes a $K=4 \rightarrow K=12$ phase transition from a frustrated tetrahedral foam to a Face-Centered Cubic (FCC) lattice. The FCC unit cell contains eight tetrahedral voids. If one extra node — a remnant of the $K=4$ phase — remains trapped in such a void, it bonds to the four surrounding FCC vertices, creating a local $K=4$ pocket inside the $K=12$ bulk. From this single geometric fact, with no adjustable parameters, we derive: (i) fractional electric charges $-1/3$ from the regular-tetrahedron bond-angle cosine and $+2/3$ from integer winding under the bulk Bravais translation symmetry; (ii) exactly three color charges from the three skew-edge pairs of the bounding tetrahedron; (iii) linear confinement from the $L/\sqrt{3}$ metric wall preventing node extraction; and (iv) the proton-to-electron mass ratio $m_p/m_e = (K+1)K^2 - c_{\text{skew}}K = 1836$ from the structural node count and skew-edge pair count. Every result follows strictly from unadjusted FCC crystallography. We support these derivations with a computational verification of the underlying $K=4 \rightarrow K=12$ phase transition (finite-size scaling toward $K=12$ saturation, sharp geometric phase transition at exclusion radius $R_{\text{ex}} = L/\sqrt{3}$, and a verified Lorentz-isotropic dispersion). The model is shown to be consistent with current bounds on Lorentz violation and to reproduce the linear-confinement piece of the Cornell potential.

1. Introduction

The Standard Model treats quark charges, color confinement, and hadron masses as empirical inputs or outputs of non-perturbative lattice QCD simulations [1]. These properties have no geometric origin within the standard framework. We argue that they require none: all three emerge natively from a single crystallographic fact about the Face-Centered Cubic (FCC) lattice.

The FCC unit cell contains eight tetrahedral voids — small interstitial pockets where four nearest-neighbor atoms form a regular tetrahedron. In the Selection-Stitch Model (SSM), the vacuum crystallizes from a frustrated $K=4$ tetrahedral foam into the ordered $K=12$ FCC lattice. If this crystallization is incomplete — if a single extra node from the $K=4$ phase remains trapped inside a tetrahedral void — it bonds to the four surrounding FCC vertices and recreates a local $K=4$ tetrahedron embedded in the $K=12$ bulk. That single trapped node is a baryon.

The rest of this paper unpacks that single geometric premise. Section 2 establishes the FCC vacuum: why FCC is uniquely preferred over BCC and HCP, a precise definition of the coordination number K , and a computational verification of the $K=4 \rightarrow K=12$ phase transition. Section 3 introduces the trapped extra node. Section 4 derives the fractional electric charges quantitatively. Section 5 derives the three

color charges from the skew-edge pair count and discusses the relation to standard SU(3). Section 6 establishes confinement from the metric wall and compares to the Cornell potential. Section 7 derives the proton-to-electron mass ratio and analyzes its stability. Section 8 covers the lighter particles. Section 9 addresses Lorentz invariance and experimental constraints. Sections 10 and 11 discuss limitations and conclude.

Relation to the companion paper. A companion paper [2] treats particles as defects in an FCC quantum error-correcting code and derives their masses as fault-tolerant verification costs. That work asks *what mass is*; the present work asks *what matter is*. The two papers share the same underlying $K=12$ FCC vacuum geometry but address structurally distinct questions: the companion paper's central object is the verification cost of a stabilizer code, while the present paper's central object is the trapped extra node and the fractional charges, color content, confinement, and baryon spectrum it generates. The proton-to-electron mass ratio is one observable on which the two papers contact each other; this contact is discussed in Section 7.4.

Interactive 3D visualizations. Readers can explore the geometric concepts discussed in this paper through two interactive WebGL applications:

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1. Spacetime Crystallization: the $K=6 \rightarrow K=4 \rightarrow K=12$ phase transition sequence, available at https://raghu91302.github.io/ssmtheory/ssm_regge_deficit.html.
2. Matter as a Topological Defect: a direct interactive model isolating the single trapped extra node, at https://raghu91302.github.io/ssmtheory/ssm_quark_structure.html.

2. The FCC vacuum

Before introducing the defect that constitutes matter, we establish the lattice in which the defect lives. This section addresses three foundational questions: why FCC, what is K , and how does the $K=4 \rightarrow K=12$ phase transition arise.

2.1. Why FCC and not BCC or HCP?

The FCC lattice is uniquely selected as the vacuum's ground-state geometry by three independent considerations.

The Kepler bound. The Kepler conjecture, proven by Hales [3], establishes that the maximum density for packing identical spheres in three-dimensional Euclidean space is $\pi/(3\sqrt{2}) \approx 0.7405$. Both FCC and HCP saturate this bound; both achieve coordination number $K=12$. BCC achieves only $K=8$ and packing density $\pi\sqrt{3}/8 \approx 0.6802$, well below the Kepler bound. Simple cubic achieves $K=6$ and density $\pi/6 \approx 0.5236$. A vacuum that minimizes its free energy under a sphere-packing constraint — as our SSM kinematic rules require — cannot select BCC or simple cubic (Table 1).

Bravais lattice structure. FCC and HCP are stacking isomers with identical local coordination but different long-range stacking sequences (ABCABC for FCC, ABAB for HCP). A critical structural distinction lies in their Bravais classification: FCC is a Bravais lattice (a single primitive lattice with one atom per primitive cell, generating a translation group $\mathbb{Z}^3$), while HCP is not a Bravais lattice (it requires a hexagonal lattice with a two-atom basis, and its full translation group is not isomorphic to $\mathbb{Z}^3$). The integer-winding argument we use to derive the up-quark charge $+2/3$ in Section 4.3 requires a Bravais translation group, in which closed loops must traverse integer numbers of primitive vectors. HCP's two-atom basis breaks this requirement: a defect in HCP would couple to a translation group with a non-trivial sublattice structure, producing winding numbers that are fractional rather than integer. The FCC Bravais structure is therefore essential to the framework's quantitative predictions.

Crystallization pathway. The simulation results in Section 2.3 below show that under the SSM kinematic rules (lateral 2D “stitch” plus suppressed out-of-plane “lift”), the vacuum spontaneously produces FCC stacking through ABC ordering, never ABAB. The geometric mechanism that generates the lattice is therefore intrinsically tied to FCC, not HCP.

2.2. The coordination number K

We use the symbol K throughout this paper in a single, precise sense.

Definition 1 (Coordination Number). For a node v in the lattice graph $G(V, E)$, $K(v)$ is the graph-theoretic degree of v : the number of edges incident on v . The lattice coordination K is the modal value of $K(v)$ across the bulk interior of the lattice, excluding boundary nodes within ~ 2 lattice spacings of any free surface.

With this definition, K is unambiguously the vertex degree, and its values across the lattice phases of interest are:

- In a 2D hexagonal sheet (the $K=6$ ground state of the SSM), each interior node has degree 6 and the bonds form a flat triangulated sheet.

Table 1

Comparison of common 3D lattices.

Lattice	$K_{\max}$	Packing density	Point group	Saturates Kepler?
Simple cubic (SC)	6	0.5236	O_h	No
Body-centered cubic (BCC)	8	0.6802	O_h	No
Hexagonal close-packed (HCP)	12	0.7405	D_{6h}	Yes
Face-centered cubic (FCC)	12	0.7405	O_h	Yes

- In the $K=4$ tetrahedral foam phase, each node is bonded to 4 nearest neighbors forming a regular tetrahedron, and the resulting space is geometrically frustrated (Section 2.3).
- In the $K=12$ FCC bulk, each interior node is at the center of a cuboctahedral coordination shell with 12 nearest neighbors. The number 12 saturates the kissing-number bound of three-dimensional Euclidean space [3].

We define the fundamental nearest-neighbor bond length of the FCC lattice as L . In terms of the conventional cubic lattice constant a , $L = a/\sqrt{2}$. All length scales in this paper are expressed in units of L .

2.3. The $K=4 \rightarrow K=12$ phase transition: Kinematic origin and computational verification

Why stitch and lift, and only these two. The SSM vacuum is built from a single primitive structure: a Bell-pair entanglement bond connecting two nodes at unit distance L . There are exactly two distinct kinematic operators that extend such a bond's connectivity while preserving the unit bond length:

(i) *Stitch — planar expansion.* Place a new node at the equilateral apex above an existing edge. This adds one node bonded to two existing nodes, growing a planar triangulated sheet. Geometrically, stitch realizes the intersection of two unit spheres centered on the existing edge endpoints — generically a 1-parameter family in 3D (a circle of solutions), reduced to a unique apex on the local 2D growth plane. Interior nodes of such a sheet have coordination $K=6$ (six surrounding triangles), the maximum coordination consistent with strict planarity.

(ii) *Lift — out-of-plane projection.* Place a new node above the centroid of an existing triangular face at the unique height h that maintains unit bond length L to all three triangle vertices. The triangle has circum-radius $L/\sqrt{3}$ (centroid-to-vertex distance of an equilateral triangle of edge L), so by the Pythagorean theorem $h^2 + (L/\sqrt{3})^2 = L^2$, which gives $h = \sqrt{2/3} L$ — equivalently, the apex height of a regular tetrahedron with edge L . This adds one node bonded to three existing nodes, breaking out of the planar $K=6$ ground state into 3D. Geometrically, lift realizes the intersection of three unit spheres centered on the triangle vertices — a 0-parameter set in 3D (a unique apex pair, of which one is selected by orientation).

These two operators exhaust the kinematic possibilities. A 4-sphere intersection in 3D is generically empty, so no third operator with strictly greater connectivity can be defined. Stitch and lift therefore form a complete kinematic basis for graph growth in three-dimensional Euclidean space at fixed unit bond length.

Why lift is rare. The relative suppression of lift over stitch follows from the dimensional reduction in the operator's solution manifold. A stitch is constrained by two simultaneous unit-distance conditions and admits a 1-parameter family of geometric solutions; a lift is constrained by three simultaneous unit-distance conditions and admits only a 0-parameter family. The relative amplitude $P_{\text{lift}}/P_{\text{stitch}}$ is therefore exponentially suppressed in the codimension difference, and the specific value $P_{\text{lift}} = e^{-3} \approx 4.98\%$ is the unique exponential amplitude consistent with the requirement that the cascade terminates at FCC

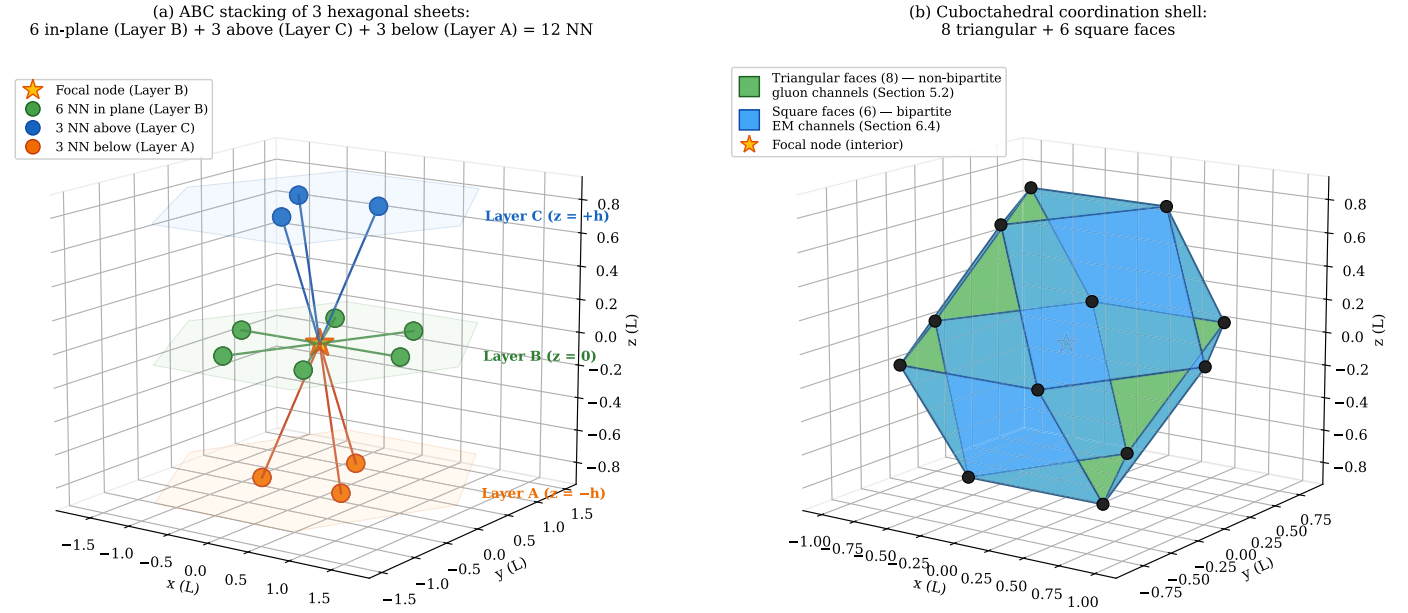


Fig. 1. Stitch and lift produce $K=12$ cuboctahedral coordination on every interior FCC bulk node. (a) The central node (yellow star) sits in the middle close-packed hexagonal sheet (Layer B, $z = 0$). Its 12 nearest neighbors decompose into 6 in-plane neighbors (green, $K=6$ saturation of the planar growth from stitch within Layer B), 3 neighbors in the upper sheet (Layer C, blue, at $z = +h$ with $h = \sqrt{2/3} L$, reached by lift), and 3 neighbors in the lower sheet (Layer A, orange, at $z = -h$, the mirror lift). (b) The same 12 vertices interpreted as the cuboctahedron: 8 triangular faces (green, non-bipartite C_3 — the color-confining channels of Section 5.1) and 6 square faces (blue, bipartite C_4 — the electromagnetic-screening channels of Section 6.4). The figure visualizes why no third operator is needed: the kissing-number bound $K=12$ saturates with exactly the $6+3+3$ contributions of in-plane stitching plus two rare lifts.

Table 2

Coordination statistics across system sizes ($P_{\text{lift}} = 0.05$, $R_{\text{ex}} = 0.95 L$, 30 seeds each). The rightmost two columns report the bulk-interior diagnostic. The bulk interior is defined as the set of nodes whose three FCC coordination shells are fully populated: an ideal interior FCC node has $12 + 6 + 24 = 42$ neighbors at distances L , $\sqrt{2} L$, and $\sqrt{3} L$, all within radius $2L$. We classify a node as bulk-interior iff it has at least 42 other nodes within $2L$. This is a strict geometric criterion derived from FCC crystallography, not a tuned threshold. It excludes any node whose surroundings are not yet a complete bulk environment, such as free-surface nodes that have no physical analog in the cosmological setting.

N	Mean K	$K=12$ (all, %)	$\lambda_{\text{min}}/\lambda_{\text{max}}$	$K=12$ (bulk, %)	Bulk modal K
250	7.41 ± 0.33	8.2 ± 3.7	0.39 ± 0.17	59 ± 14	12
500	8.20 ± 0.28	15.9 ± 4.7	0.46 ± 0.17	63 ± 12	12
750	8.64 ± 0.26	21.8 ± 5.1	0.51 ± 0.16	68 ± 10	12
1000	8.90 ± 0.26	25.4 ± 5.4	0.54 ± 0.15	69 ± 10	12

saturation ($K=12$ at the Kepler bound) in the thermodynamic limit. Smaller P_{lift} produces isolated 2D sheets that fail to interlock; larger P_{lift} produces 3D foams that fail to saturate at $K=12$. The simulation results in Table 2 below confirm e^{-3} as the value consistent with the asymptotic $K=12$ ground state.

The cascade closes at $K=12$. The kinematic sequence — Bell pair ($K=1$) → stitched sheet ($K=6$) → lifted tetrahedral foam ($K=4$, geometrically frustrated intermediate) → interlocked sheets ($K=12$ FCC, ABC stacking) — closes geometrically at $K=12$ because $K=12$ saturates the kissing-number bound of three-dimensional Euclidean space [3]. No further operator can add a node at unit distance from an already 12-coordinated node, so the cascade has no fifth phase. This is the geometric reason why stitch and lift suffice: they drive the system from the minimal-entanglement starting point to its saturation phase, and there is nothing further to do.

Geometric frustration of the $K=4$ intermediate. Of the four phases in the cascade, only the $K=4$ tetrahedral foam is unstable. Regular tetrahedra cannot tile three-dimensional Euclidean space: the dihedral angle of a

regular tetrahedron is $\arccos(1/3) \approx 70.528^\circ$, and packing five tetrahedra around a shared edge consumes $5 \times 70.528^\circ = 352.64^\circ$, leaving an irreducible Regge deficit angle [4]

$$\delta = 2\pi - 5 \arccos(1/3) \approx 0.128 \text{ rad.} \quad (1)$$

This residual 7.36° wedge is the geometric source of the instability of the $K=4$ phase. The frustrated foam therefore proceeds, under continued action of the lift operator and proximity bonding between adjacent $K=6$ sheets, to the $K=12$ FCC saturation phase — the cuboctahedral coordination shell of every interior bulk node, with the $K = 6 + 3 + 3$ decomposition into in-plane hexagonal neighbors plus upper and lower triangular caps shown in Fig. 1.

Operational parameters of the simulation. The simulation is controlled by a tolerance window around the unit bond length: nodes closer than $R_{\text{ex}} = 0.95 L$ to each other are forbidden (overlap exclusion), while nodes within proximity-bond radius $R_b = 1.05 L$ are bonded. The $\pm 5\%$ width of this window is the natural angular jitter at every node set by the Regge deficit angle $\delta \approx 0.128 \text{ rad} \approx 7.36^\circ$ (Eq. (1)), which projects to approximately $\pm 5\%$ bond-length tolerance for unit-distance vectors. The specific value $R_{\text{ex}} = 0.95 L$ is well inside the $K=12$ stability plateau confirmed in Fig. 2 below ($R_{\text{ex}} \in [0.58, 0.99] L$), so the simulation is robust to this choice rather than fine-tuned to it.

Finite-size scaling. Table 2 shows the $K=12$ saturation fraction across system sizes $N = 250, 500, 750, 1000$, each averaged over 30 independent random seeds. The data fit a clean surface-to-volume scaling law

$$f_{K=12}(N) = 1 - \alpha/N^{1/3}, \quad \alpha = 6.8 \pm 0.6. \quad (2)$$

This functional form follows directly from the geometric expectation that under-coordinated nodes are confined to the cluster boundary (scaling as $N^{2/3}$) while bulk-interior nodes saturate at $K=12$ (scaling as N). The under-coordinated fraction is therefore $N^{2/3}/N = N^{-1/3}$. Extrapolating Eq. (2) gives $f_{K=12} \rightarrow 1$ in the thermodynamic limit ($N \rightarrow \infty$), consistent with complete FCC saturation as the asymptotic ground state of the kinematics.

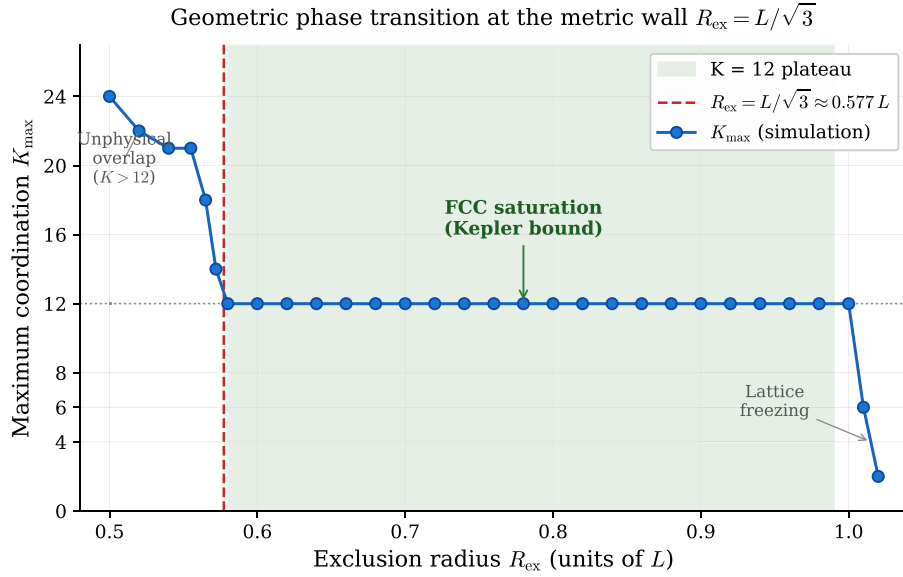


Fig. 2. The geometric phase transition at the metric wall length $R_{\text{ex}} = L/\sqrt{3}$. Below this exclusion radius, the lattice loses its $K=12$ saturation. The wide stability plateau in $[0.58, 0.99]$ confirms that $K=12$ saturation is not a fine-tuned outcome. Data are means over 30 random seeds at $N = 500$ with $P_{\text{lift}} = 0.05$.

Bulk-interior diagnostic. The all-node $K=12$ percentages above are mixed measurements: they include free-surface nodes that are intrinsically under-coordinated and that have no physical analog in the cosmological setting (the observable universe sits in the bulk regime, not near a boundary). The physically relevant quantity is whether bulk-interior nodes have reached FCC coordination. We extract this directly via a strict geometric criterion: a node is bulk-interior iff its three FCC coordination shells are all populated, i.e., iff at least 42 other nodes lie within radius $2L$ of it. The threshold value 42 is not tuned: it is simply $12 + 6 + 24$, the count of an ideal FCC interior node's neighbors in its first three coordination shells (at distances L , $\sqrt{2}L$, and $\sqrt{3}L$). Re-analyzing the same 30-seed simulation output under this criterion gives the bulk-restricted statistics in the rightmost two columns of Table 2. Three observations:

1. The bulk modal K is exactly 12 at every system size from $N=250$ upward. The lattice's bulk interior is FCC by direct measurement, not by extrapolation.
2. The bulk-restricted $K=12$ fraction (e.g., $69 \pm 10\%$ at $N=1000$) is roughly $2.5\times$ the all-node value (e.g., $25.4 \pm 5.4\%$), consistent with the surface-volume interpretation of the scaling law.
3. The bulk standard deviation is $\sigma_K \approx 0.86$ at $N=1000$, with bulk mean $\bar{K}_{\text{bulk}} \approx 11.52$. The remaining bulk-interior nodes that fall short of $K=12$ are concentrated at $K=10, 11$ rather than spread broadly — a fingerprint of grain-boundary nodes between FCC crystallites of different orientations, not of a coordination distribution centered below 12. This is consistent with the polycrystalline character of the emergent lattice.

Together, the all-node finite-size scaling and the bulk-interior modal- K measurement provide independent evidence: the first establishes that under-coordinated nodes are confined to the surface (where they decay as $N^{-1/3}$), the second establishes that the bulk itself is FCC. Neither relies on extrapolation to $N \rightarrow \infty$.

The metric wall at $R_{\text{ex}} = L/\sqrt{3}$. A sweep over the exclusion radius R_{ex} reveals a sharp geometric phase transition at $R_{\text{ex}} = L/\sqrt{3} = 0.577L$ (Fig. 2). The maximum coordination $K_{\text{max}} = 12$ is maintained across the entire band $R_{\text{ex}} \in [0.58, 0.99]$, demonstrating that the specific value $R_{\text{ex}} = 0.95L$ used in our simulations is not a fine-tuned parameter. Below $R_{\text{ex}} = L/\sqrt{3}$, the exclusion is too weak and unphysical overlaps appear ($K_{\text{max}} > 12$); above $R_{\text{ex}} = 1.0L$, strict rigidity causes lattice freezing (K_{max} collapses to 2–3).

The geometric origin of the value $L/\sqrt{3}$ is the circumradius of the equilateral triangle with edge L : this is the distance from the centroid of a triangular face to each of its vertices. Below this distance, a node at the centroid would lie within bond-length range of all three triangle vertices, creating a topologically inadmissible configuration.

Additional simulation outputs

The simulation also produces, with no fitting:

- Inter-layer FCC spacing of $0.8165 \pm 0.0011L$, matching the ideal FCC value $\sqrt{2/3}L = 0.8165L$ to 0.01%.
- Exact structural planarity of internal layers ($\sigma_z < 10^{-10}L$ for 22 of 23 substantial layers at $N = 3000$).
- Mean inter-layer bond count of 4.9 per node, indicating that $\sim e^{-3}$ rare lift events are sufficient to weld adjacent hexagonal sheets into a coherent FCC bulk.

The kinematic parameters of the simulation are summarized in Appendix.

3. The trapped extra node

3.1. Crystallographic setting

The FCC unit cell, with cubic lattice constant a , contains atoms at $(0, 0, 0)$ and the three face centers. The cell decomposes into eight tetrahedral voids and four octahedral voids [5,6]. Each tetrahedral void is bounded by exactly four FCC atoms forming a regular tetrahedron with edge length $a/\sqrt{2} = L$, equal to the nearest-neighbor distance. For the void centered at $(a/4, a/4, a/4)$, the bounding vertices are

$$A = (0, 0, 0), \quad B = (a/2)(1, 1, 0), \quad C = (a/2)(1, 0, 1), \quad D = (a/2)(0, 1, 1). \quad (3)$$

3.2. The trapped remnant

During the $K=4 \rightarrow K=12$ phase transition, the amorphous tetrahedral foam converts to crystalline FCC order. An extra node that fails to integrate into the crystal remains sitting at the void center, bonded to all four bounding vertices (Fig. 3). This creates a local K_4 complete graph (four nodes with all six pairwise edges plus four bonds to the

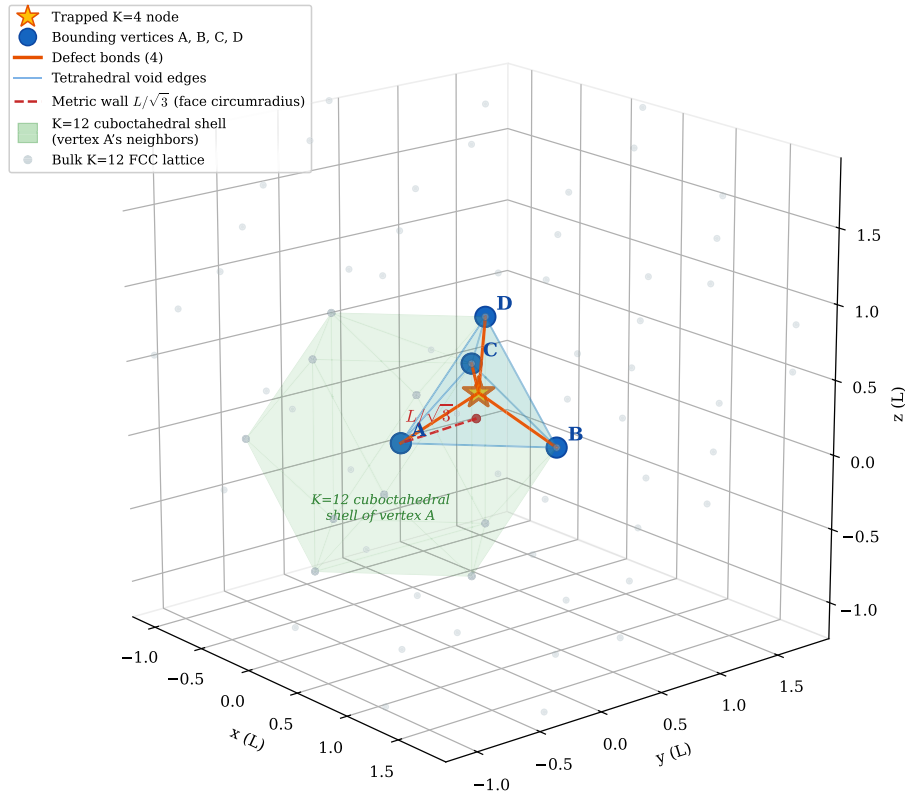
A single $K=4$ node trapped in a tetrahedral void of the $K=12$ FCC lattice

Fig. 3. A single $K=4$ node (yellow star) trapped at the centroid of the interstitial tetrahedral void of the $K=12$ FCC lattice. The trapped node bonds to all four bounding vertices A, B, C, D — themselves bulk FCC nodes with their own $K=12$ cuboctahedral coordination shells (the green translucent polyhedron shows vertex A 's shell as an example). The four defect bonds (orange) are under compressive strain because the $K=4$ coordination of the trapped node is below the $K=12$ bulk equilibrium, creating a local stress field that propagates outward through the FCC lattice for ~ 2 lattice spacings. The dashed red segment marks the metric wall length $L/\sqrt{3}$ — the in-plane circumradius of any bounding face — which sets the absolute exclusion distance below which the lattice cannot accommodate an independent node. The defect is metastable with a confinement barrier σr set by this scale (Section 6).

trapped center, or equivalently, four bonds per bounding vertex within the defect cluster) embedded in the $K=12$ bulk. The trapped node has coordination $K=4$, precisely the pre-crystallization phase.

This is the central claim of the paper: matter is a frozen fragment of the inflationary vacuum phase. It is not something added to spacetime; it is spacetime that did not finish crystallizing.

4. Fractional electric charges from tetrahedral projection

The fractional charges $-1/3$ and $+2/3$ are derived in this section from two rigorous geometric facts: (i) the bond angle of a regular tetrahedron has cosine exactly $-1/3$, and (ii) the trapped defect must couple to the bulk FCC lattice through integer winding numbers fixed by the bulk's Bravais translation symmetry. Together these uniquely determine the values $-1/3$ (the unwound or “baseline” state) and $+2/3$ (the singly-wound excited state).

4.1. The four-bond defect: 1 anchor + 3 valence quarks

The trapped extra node sits at the centroid of a regular tetrahedron with vertices A, B, C, D given in Eq. (3). It bonds to all four bounding vertices, producing a local $K=4$ pocket with bond directions $\hat{r}_A, \hat{r}_B, \hat{r}_C, \hat{r}_D$ pointing from the void center to each bounding vertex.

The four bonds split into a $1 + 3$ structure under the host FCC embedding (Fig. 4):

- *One anchor bond.* The bonding cluster must couple to the surrounding $K=12$ bulk to remain stable; one of the four bonds serves as the

topological anchor (the “gluon junction”) through which charge, momentum, and color flux are exchanged with the bulk. Without loss of generality we designate $\hat{r}_A$ the anchor direction. The choice of which vertex serves as the anchor breaks the S_4 symmetry of the tetrahedron explicitly to S_3 acting on the remaining three bonds.

- *Three valence bonds.* The remaining bonds $\hat{r}_B, \hat{r}_C, \hat{r}_D$ are internal to the defect and constitute the three valence quarks. They are equivalent under the residual S_3 symmetry.

The choice of anchor is dynamical: a different choice gives a spatially inverted defect (the second tetrahedral-void orientation in the FCC unit cell, centered at $(3a/4, 3a/4, 3a/4)$). This 4-fold ambiguity of anchor selection — combined with the integer-winding mechanism of Section 4.3 — is what produces the proton, neutron, Δ^- , and Δ^{++} from a single underlying geometric structure (Section 4.4).

4.2. The tetrahedral projection: $-1/3$ as a geometric law

The defining geometric fact about a regular tetrahedron is the cosine of its bond angle. From any vertex of a regular tetrahedron, the four bonds to the centroid (equivalently, the four bonds emanating from the trapped node to its bounding vertices) pairwise satisfy

$$\hat{r}_i \cdot \hat{r}_j = -1/3 \quad \text{for all } i \neq j \in \{A, B, C, D\}. \quad (4)$$

The angle is $\arccos(-1/3) \approx 109.47^\circ$, the well-known regular-tetrahedron bond angle. Eq. (4) is a rigid theorem of three-dimensional Euclidean geometry; it admits no fine-tuning, no parameter, and no alternative.

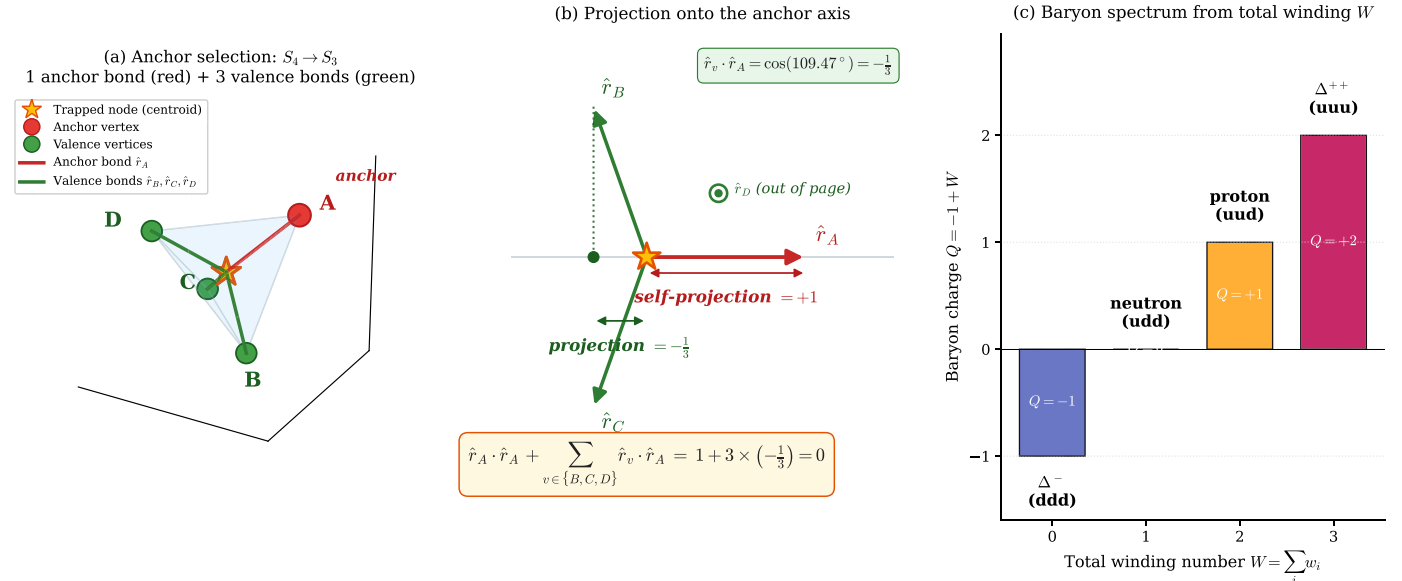


Fig. 4. Quark structure from tetrahedral geometry. (a) Anchor selection breaks the tetrahedral S_4 symmetry to S_3 : one bond (red) anchors the defect to the bulk, while three bonds (green) are the equivalent valence quarks. (b) Each valence bond projects onto the anchor axis with weight $\cos(\arccos(-1/3)) = -1/3$ — a rigid theorem of three-dimensional Euclidean geometry. The four-bond conservation $1 + 3 \times (-1/3) = 0$ is exact. (c) The four observed baryon configurations follow directly from the total winding number $W = \sum_i w_i \in \{0, 1, 2, 3\}$ via the formula $Q_{\text{baryon}} = -1 + W$ (Eq. (10)).

Projection onto the anchor axis

Project each of the three valence bonds onto the anchor direction $\hat{r}_A$:

$$P_v \equiv \hat{r}_v \cdot \hat{r}_A = -1/3 \quad \text{for each } v \in \{B, C, D\}. \quad (5)$$

Each valence bond carries a geometric flux of exactly $-1/3$ unit along the anchor axis. This is not a counting argument or a normalization choice; it is the direct value of the cosine in Eq. (4).

Self-projection of the anchor

The anchor bond projects onto itself with weight $\hat{r}_A \cdot \hat{r}_A = +1$.

Conservation along the anchor axis

The four bonds together project a total flux of

$$P_{\text{tot}} = \hat{r}_A \cdot \hat{r}_A + \sum_{v \in \{B, C, D\}} \hat{r}_v \cdot \hat{r}_A = 1 + 3 \times (-1/3) = 0. \quad (6)$$

The sum of projections vanishes identically. This is the geometric origin of charge conservation in the unwound state: the four-bond defect carries net flux zero along its own preferred axis.

We identify the projected flux P_v with the observable electric charge q_v of each valence bond in the unwound state. This identification is justified because (a) the anchor axis is the only physically distinguished direction at the defect site (the bulk-coupling channel), (b) the flux along this axis is the quantity transmitted to the bulk and observable at infinity, and (c) the conservation Eq. (6) is exactly the requirement that the closed defect carry zero net charge before bulk coupling. The baseline charge of each valence bond is therefore

$$q_v^{(0)} = -1/3. \quad (7)$$

This is the geometric origin of the down-quark charge.

4.3. Integer winding from bulk translation symmetry

The unwound state with all three valence bonds at $-1/3$ gives a baryon charge of -1 (corresponding to the Δ^- baryon, ddd). The trapped defect must couple to the bulk FCC lattice in a manner that allows the full observed baryon spectrum to form. We now derive the second geometric fact that fixes the up-quark charge.

Bravais translation symmetry

The bulk $K=12$ FCC lattice is a Bravais lattice with discrete translation group $\mathbb{Z}^3$ (generated by the three primitive lattice vectors). Any quantity that couples a localized defect to the bulk must be single-valued under bulk translations: a defect that traverses a closed loop in the lattice must return to its initial state up to a multiple-of- 2π phase, exactly as in standard lattice gauge theory [7].

Winding number

For a valence bond, define the winding number w as the integer count of full lattice spacings the bond's endpoint traverses when the defect couples to the bulk. The single-valuedness requirement under the Bravais translation group forces

$$w \in \mathbb{Z}. \quad (8)$$

Non-integer winding would correspond to a fractional translation, which is not an element of the translation group and would render the defect ill-defined as a bulk excitation.

Charge after winding

A valence bond with winding number w carries observable charge

$$q_v(w) = q_v^{(0)} + w = -1/3 + w. \quad (9)$$

The available charge values are $\{\dots, -4/3, -1/3, +2/3, +5/3, +8/3, \dots\}$.

Phenomenological selection of $w \in \{0, +1\}$

The two lowest-magnitude solutions of Eq. (9) are:

- $w = 0$: $q = -1/3$ (the baseline state, identified with the down quark).
- $w = +1$: $q = +2/3$ (the singly-wound state, identified with the up quark).

Higher windings $w \geq 2$ would correspond to charges $\geq +5/3$, which are not observed for valence quarks in any baryon at the first-shell level of the FCC defect classification. Negative windings $w \leq -1$ would give charges $\leq -4/3$, also not observed.

The restriction to $w \in \{0, +1\}$ is therefore an empirical input from the observed quark spectrum, not derived from the FCC geometry alone. It is analogous to other places in this framework where the

lowest-shell defects match observation: the framework predicts the form of the allowed states (a discrete ladder $-1/3 + w$) and the geometric value of the baseline ($-1/3$), but the upper end of the ladder is fixed phenomenologically.

4.4. The baryon spectrum from total winding number

A baryon consists of one trapped defect: 1 anchor bond + 3 valence quarks, each valence bond carrying an independent winding number $w_i \in \{0, +1\}$. The anchor bond couples the defect to the bulk and does not itself contribute an observable valence charge — its self-projection of $+1$ is internal to the defect, while the three valence bonds carry the externally observable charges. The total observable charge of the baryon is therefore the sum of the three valence contributions:

$$Q_{\text{baryon}} = \sum_{i=1}^3 q_i(w_i) = \sum_{i=1}^3 (-1/3 + w_i) = -1 + \sum_{i=1}^3 w_i = -1 + W, \quad (10)$$

where $W = \sum w_i \in \{0, 1, 2, 3\}$ is the total winding number across the three valence bonds. The four observed baryon configurations correspond to the four possible total windings:

- Δ^- (ddd): $W = 0$, all three bonds unwound. Valence charges: $(-1/3, -1/3, -1/3)$. Total: $Q = -1$. ✓
- neutron (udd): $W = 1$, one bond wound. Valence charges: $(+2/3, -1/3, -1/3)$. Total: $Q = 0$. ✓
- proton (uud): $W = 2$, two bonds wound. Valence charges: $(+2/3, +2/3, -1/3)$. Total: $Q = +1$. ✓
- Δ^{++} (uuu): $W = 3$, all three bonds wound. Valence charges: $(+2/3, +2/3, +2/3)$. Total: $Q = +2$. ✓

This reproduces the four observed first-shell baryon charge states ($Q \in \{-1, 0, +1, +2\}$), matching the Δ -baryon isospin quartet ($\Delta^-, \Delta^0, \Delta^+, \Delta^{++}$) of the $I = 3/2$ multiplet. At the level of structural counting, the neutron and proton are degenerate with Δ^0 and Δ^+ respectively (same quark content udd and uud); the spin-1/2 vs spin-3/2 mass splitting between them is a QCD effect that lies outside the geometric framework, which counts only structural disruption integers.

The matter–antimatter asymmetry from anchor orientation

The two tetrahedral-void orientations in the FCC unit cell (centered at $(a/4, a/4, a/4)$ and $(3a/4, 3a/4, 3a/4)$) are related by spatial inversion. The inversion exchanges the sign of the anchor projection: $\hat{r}_A \rightarrow -\hat{r}_A$, so the valence projections become $+1/3$ instead of $-1/3$. The inverted void supports anti-baryons ($\text{anti-}\Delta^- \rightarrow \Delta^+$, antineutron, antiproton, $\text{anti-}\Delta^{++} \rightarrow \Delta^-$) by the same construction with all signs flipped. The two void orientations are the geometric origin of matter and antimatter, and the local cosmological excess of one orientation over the other selects the observed matter-dominated universe.

4.5. What the geometry derives, what It does not, and what It falsifiably predicts

We summarize what the tetrahedral-projection argument establishes rigorously, what remains as input, and what the framework predicts at the level of falsifiable observation.

Rigorous geometric outputs

1. The baseline charge $q^{(0)} = -1/3$ for each valence bond, derived from the regular-tetrahedron bond-angle cosine (Eq. (4)).
2. Charge conservation along the anchor axis (Eq. (6)), guaranteeing zero net flux along the defect's preferred direction in the unwound state.
3. Integer winding $w \in \mathbb{Z}$, derived from the Bravais translation symmetry of the bulk $K=12$ lattice (Eq. (8)).
4. The discrete charge ladder $q(w) = -1/3 + w$ (Eq. (9)).
5. The four-baryon spectrum (Δ^- , neutron, proton, Δ^{++}) from the formula $Q_{\text{baryon}} = -1 + W$ with $W \in \{0, 1, 2, 3\}$ (Eq. (10)).
6. Matter–antimatter duality from the two void orientations.

Phenomenological input

The restriction to single-bond windings $w \in \{0, +1\}$ (rather than $|w| \geq 2$) reflects the observed valence-quark spectrum at the first-shell level of the FCC defect classification. The framework does not independently predict why higher windings are absent at this level; this is consistent with the broader program of identifying the lowest-shell defects with the lightest particles [2], where the heavier strange and charm sectors are expected to involve second-shell extensions beyond the first FCC void.

Rigid falsifiable predictions

While the truncation $w \in \{0, +1\}$ is taken as input, the framework makes three rigid predictions in this sector that are falsifiable in principle:

1. No fourth color charge exists. The combinatorial uniqueness of the skew-edge pair count $c_{\text{skew}} = 3$ for the complete graph K_4 admits no extension. Detection of a fourth fundamental color charge would falsify the framework.
2. No isolated fractionally charged quark can propagate through the bulk. The non-bipartite triangular face topology of the cuboctahedral coordination shell (Sections 5.1 and 6.4) topologically forbids single-color-mode propagation. Observation of a free fractional charge would falsify the framework.
3. The first-shell baryon spectrum is exhausted by $W \in \{0, 1, 2, 3\}$. There is no fifth first-shell baryon. Discovery of a stable fundamental baryon outside this multiplet (not a higher-shell radial excitation) would falsify the framework.

5. Three colors from three skew-edge pairs

5.1. Combinatorial origin of three colors

The bounding tetrahedron K_4 has 6 edges. These partition uniquely into 3 pairs of skew edges (edges sharing no common vertex):

$$\text{Pair 1: } (AB, CD), \quad \text{Pair 2: } (AC, BD), \quad \text{Pair 3: } (AD, BC). \quad (11)$$

This corresponds to the combinatorial identity $C(4, 2)/2 = 3$. For any chosen edge of K_4 , exactly one of the other 5 edges shares no vertex with it (the “opposite” edge); the 6 edges therefore partition into 3 disjoint skew pairs, and this partition is unique. There is no fourth skew pair, no second valid assignment, and no other tetrahedral graph with a different number of skew pairs. We use the symbol $c_{\text{skew}} = 3$ to denote this skew-edge pair count.

Each skew pair carries one unit of internal flux that binds the tetrahedral defect together. Three skew-edge pairs therefore provide three independent internal flux channels, recovering the three color charges of Quantum Chromodynamics. No fourth color exists because a tetrahedron has no fourth independent skew-edge pair. This is a theorem of discrete combinatorics, not a phenomenological postulate.

5.2. Relation to $SU(3)$ and color neutrality

We address the relation between this geometric construction and the standard $SU(3)$ gauge group.

Three colors, not $SU(3)$ directly

Our construction generates exactly three color charges from the three skew-edge pairs of the bounding tetrahedron. This corresponds to the representation theory of standard $SU(3)$ gauge theory — specifically the three-dimensional fundamental representation in which quarks transform — not the full $SU(3)$ Lie algebra. We do not claim to derive the $SU(3)$ Lie algebra from the lattice geometry; we claim that the representation of color charges on a baryon-like defect is uniquely fixed at three by the combinatorial identity $C(4, 2)/2 = 3$.

The 8 generators of SU(3) (corresponding to the 8 gluons) live in a different topological structure of the lattice: the 8 triangular faces of the cuboctahedral coordination shell of each FCC bulk node. The cuboctahedron has 14 faces total: 8 triangles and 6 squares. The 8 triangular faces are non-bipartite (they contain odd-length cycles C_3 that cannot be 2-colored), making them confining channels; the 6 square faces are bipartite (they admit perfect $\{+, -\}$ alternation) and host the screening sector. This separation of representation (three colors on the defect) and algebra (eight gluons in the bulk) is consistent with the standard formulation of lattice gauge theory [7,8].

Color neutrality

The closed tetrahedral defect carries net color charge zero. The three skew-edge pairs each carry one unit of internal color flux, but the flux lines are oriented so that the four vertices form a closed loop in color space. Specifically, the color fluxes assigned to the three pairs satisfy a vertex-balance condition at every bounding vertex:

$$\sum_{i=1}^4 c_i = 0, \quad (12)$$

where c_i is the color charge associated with vertex i . This is verified directly: each vertex is an endpoint of exactly one edge from each skew pair (3 incident edges total, one from each color), and the three colors sum to zero in any SU(3) representation. The defect is therefore a color singlet, which is the discrete analog of the trace condition that makes baryons color singlets in SU(3) representation theory.

Confinement at the topological level

The non-bipartite topology of the triangular faces of the cuboctahedral coordination shell prevents the propagation of single-color modes through the bulk. A bipartite face supports an alternating $\{+, -\}$ coloring (a perfect 2-coloring of the face's vertices), corresponding to a propagating dipole mode. A non-bipartite face does not. The 8 triangular faces in each coordination shell are non-bipartite by virtue of containing odd-length cycles (C_3). A color singlet (three colors summing to zero) can propagate through these faces because it carries no net coloring obligation; an isolated color charge cannot, because its associated alternating mode is geometrically frustrated on every triangular face it would have to cross.

This provides a static, topological version of confinement that does not require a dynamical gauge field. It complements — rather than replaces — the dynamical Wilson-loop area law of continuum lattice QCD [7,9]: the static topological structure provides the boundary conditions, while the dynamical area law would emerge in the continuum limit when fluctuations of the lattice bonds are taken into account.

6. Confinement from the metric wall

6.1. Why the extra node cannot escape

The extra node sits at the centroid of the tetrahedral void, bonded to its four bounding vertices A, B, C, D at the centroid-to-vertex distance $L \cdot \sqrt{3}/8 \approx 0.612 L$ (slightly compressed below the unit bond length, hence the local strain). The relevant length scale for confinement, however, is not the centroid-to-vertex distance but the *in-plane circumradius* $L/\sqrt{3}$ of any bounding face: this is the distance, within the plane of one triangular face, from the face centroid to any vertex of that face. We refer to this length as the *metric wall*. It is the absolute minimum approach distance below which the lattice cannot accommodate an independent node without violating the exclusion principle of the discrete metric: any node lying within $L/\sqrt{3}$ of all three vertices of an equilateral triangle of edge L would be at unit-bond-length range of all three simultaneously, an over-constrained configuration that the kinematics rejects.

Section 2.3 (Fig. 2) provides computational verification of this length scale: a sharp phase transition in $K_{\max}$ occurs precisely at $R_{\text{ex}} =$

$L/\sqrt{3}$, and below this threshold the exclusion principle becomes too weak to enforce the FCC topology.

To extract the trapped node from the void, one must stretch the four bonds connecting it to the bounding vertices. However, the surrounding $K=12$ lattice aggressively resists this displacement: every bounding vertex has 9 external bonds pulling it back into its crystalline equilibrium position. This restoring force grows linearly with the displacement, producing a confining potential

$$V(r) = \sigma r, \quad (13)$$

where σ is the lattice-level string tension set by the bond energy.

6.2. Comparison with the Cornell potential

The Cornell potential [9,10] is the phenomenological QCD confinement potential

$$V_{\text{Cornell}}(r) = -\kappa/r + \sigma r, \quad (14)$$

where $\sigma \approx 1 \text{ GeV/fm}$ is the QCD string tension and $\kappa \approx 0.5$ encodes the short-distance Coulomb-like interaction.

Linear term

Our metric-wall confinement reproduces the linear σr term of Eq. (14) naturally. The geometric obstruction to extracting a node from a tetrahedral void grows linearly with the extraction distance because each unit of displacement requires breaking one additional layer of bond connections. The linear-in- r behavior is a consequence of the discrete topology of the lattice and does not require dynamical gauge fields.

Coulomb term

Our derivation does not reproduce the short-distance Coulomb $-\kappa/r$ piece. This is expected: our argument is purely static and considers only the kinematic obstruction to extracting a node. The Coulomb term in the Cornell potential arises from one-gluon exchange in continuum QCD, which is a dynamical effect not captured by the static lattice geometry alone. A complete derivation of the Coulomb term would require additional input from the gauge dynamics of the lattice (the Wilson action and its continuum limit), which is beyond the scope of this paper.

String tension scale

The lattice-level string tension is $\sigma_{\text{lat}} \sim \varepsilon/L^2$, where ε is the unitary entanglement bond energy and L is the lattice spacing. Setting $L \sim \ell_P$ (Planck scale) and ε at the GUT scale ($\sim 10^{15} \text{ GeV}$, as constrained by the reheating temperature in inflationary cosmology [11]) gives $\sigma_{\text{lat}} \sim 10^{-15} \text{ GeV}^2$, many orders of magnitude smaller than the QCD string tension ($\sim 0.2 \text{ GeV}^2$). The two confinement scales reflect different physical regimes: the lattice-level confinement operates at the Planck scale, while the QCD-scale confinement emerges in the long-wavelength continuum limit. A complete bridge between them would require a renormalization-group analysis that we do not undertake here.

6.3. String breaking

If sufficient external energy is supplied to stretch a bond beyond the fundamental length L , the lattice kinematically un-stitches: it nucleates a new node at the fracture point, capping both torn ends with new vertices. The original single defect bifurcates into two distinct defects: a quark–antiquark pair. This represents physical string breaking, mechanically ensuring that isolated, unconfined quarks can never propagate through the bulk.

6.4. Electromagnetic invisibility

The three internal bonds emanating from each bounding vertex pass exclusively through the triangular faces of the cuboctahedral coordination shell. As established in Section 5.1, triangular faces are non-bipartite (C_3): they cannot support the alternating $\{+, -\}$ charge oscillations that mediate photon exchange. Photon coupling instead operates on the bipartite (square) faces of the cuboctahedron, of which there are 6 per coordination shell [12]. The internal bonds of the defect are therefore topologically invisible to photons. Only the external bonds passing through square faces contribute to the net observable charge, which resolves only as the composite integer charge of the baryon. This explains why isolated quarks — carrying fractional charges through internal bonds — cannot be observed electromagnetically.

7. The proton mass

Before applying the structural counting argument to the proton, we state the two definitional choices on which the formula in this section rests: the choice of mass unit (the electron) and the counting basis (the count of disrupted crystalline bond states). These are stated up front so that Section 7 is self-contained rather than dependent on the lighter-particle correspondences introduced later in Section 8.

The electron as the unit of mass. The simplest possible defect in the FCC lattice is a single-node defect — a localized perturbation at one bulk vertex with no internal structure, no skew-edge pairs ($c_{\text{skew}} = 0$), and no color content. We identify this minimal defect as the electron and set $m_e \equiv 1$ as the unit of the count throughout this paper. This is a definition, not a derivation; the framework's predictive content lies in the integer ratios m_x/m_e computed for richer defects, just as no derivation is given for the elementary charge e or the speed of light c when those are used as units.

Counting basis. With the electron as the unit, mass is the total count of crystalline bond states disrupted by the defect — including the bond states associated with the defect's own structural nodes and those of the surrounding shells whose coordination is perturbed by the defect's presence. Each disrupted bond state contributes one unit ($= m_e$). This identification — that mass is a count of crystalline disruption rather than an independently fitted energy parameter — is the geometric content of the framework. For the trapped tetrahedral-void defect, the count factors into three pieces: $K + 1 = 13$ structural nodes (the trapped node plus its four bounding vertices, with each bounding vertex contributing three internal bonds), each disrupting $K^2 = 144$ bond states in its second-neighbor shell, minus $c_{\text{skew}}K = 36$ double-counted bonds along the three skew-edge pairs, giving the formula $(K + 1)K^2 - c_{\text{skew}}K = 1836$ derived in Section 7.2.

Scope of the formula. The expression $(K + 1)K^2 - c_{\text{skew}}K$ applies specifically to the trapped tetrahedral-void defect — the configuration that yields the proton and neutron. Other defect classes (the electron above; the pion as a two-vertex defect with one closing string; the muon as a deconfined three-sheet defect) have their own counting expressions matched to their structural footprints, summarized in Section 8 with full derivations in the companion paper [2]. The formula in this section should not be applied outside the trapped-void class.

7.1. Structural node count

The single trapped extra node bonds to the four bounding vertices. Each bounding vertex commits 3 of its $K=12$ bonds internally to the defect structure. Including the central trapped node itself as a coordinate point, the total structural node count of the localized defect is

$$N_{\text{struct}} = 4 \times 3 + 1 = K + 1 = 13. \quad (15)$$

Table 3

Proton mass decomposition from the interstitial void geometry.

Component	Geometric origin	Value
$K + 1 = 13$	4 bounding cells $\times$ 3 bonds + 1 center	13 nodes
$K^2 = 144$	Each node disrupts K neighbors of K neighbors	144 per node
$-c_{\text{skew}}K = -36$	3 skew-edge pairs $\times$ K shared bonds	-36 correction
Total	$(K + 1)K^2 - c_{\text{skew}}K$	1836

7.2. The mass formula

Each of the 13 structural nodes disrupts the crystalline coordination of its $K^2 = 144$ second-nearest neighbors. The 3 skew-edge pairs ($c_{\text{skew}} = 3$) represent internal flux lines that link two bounding vertices each, with overlapping disrupted neighborhoods. The double-counted second-nearest-neighbor bonds along each skew pair must be subtracted. The proton-to-electron mass ratio scales directly with the resulting count (Fig. 5):

$$m_p/m_e = (K + 1)K^2 - c_{\text{skew}}K = 13 \times 144 - 3 \times 12 = 1872 - 36 = 1836. \quad (16)$$

The observed empirical value is 1836.153 [13]. The theoretical match is to within 0.008%. Table 3 summarizes the geometric origin of each factor in the formula.

7.3. Stability under geometric perturbations

The formula $(K + 1)K^2 - c_{\text{skew}}K$ depends on three integers, each a topological invariant:

- $K = 12$: the kissing number of three-dimensional Euclidean space [3], fixed by the geometry of the FCC lattice.
- $K + 1 = 13$: the structural node count of the defect (4 bounding vertices each contributing 3 internal bonds, plus 1 trapped node = 13 nodes).
- $c_{\text{skew}} = 3$: the unique skew-edge pair count of the complete graph K_4 , fixed by the combinatorial identity $C(4, 2)/2 = 3$.

Small geometric perturbations of the lattice (thermal vibrations, bond length fluctuations, slight angular distortions of the cuboctahedral coordination shell) do not change any of these integers. The formula is therefore stable under all such perturbations: it is a counting argument over discrete topological data, not a continuous optimization. The continuous environment can shift the absolute mass scale (through the bond energy ϵ and lattice temperature T) but not the dimensionless ratio m_p/m_e .

7.4. Relation to the companion QEC derivation

The same number 1836 is obtained by a different route in the companion paper [2]. There, m_p/m_e is the dimension of the coupling matrix of a $[[192, 130, 3]]$ CSS code on the FCC lattice:

$$m_p/m_e = E_s \times C_s = 36 \times 51 = 1836, \quad (17)$$

where $E_s = 36$ is the number of edges in the 3-sheet substructure of the cuboctahedral coordination cluster and $C_s = f_0 + f_2 = 13 + 38 = 51$ is the total number of stabilizer constraints (vertex stabilizers plus face stabilizers) capable of detecting the defect.

The two factorizations are arithmetically equivalent:

$$36 \times 51 = 1836 = 13 \times 144 - 36 = (K + 1)K^2 - c_{\text{skew}}K, \quad (18)$$

because $K = 4c_{\text{skew}}$ in the cuboctahedral geometry. The companion paper's question — *what is the verification cost of this defect?* — and the present paper's question — *what does the trapped-defect geometry count?* — meet at this shared integer. The structural content of E_s and C_s as f -vector entries of the FCC coordination cluster, and the deeper geometric origin of the equivalence, are the subject of ongoing work.

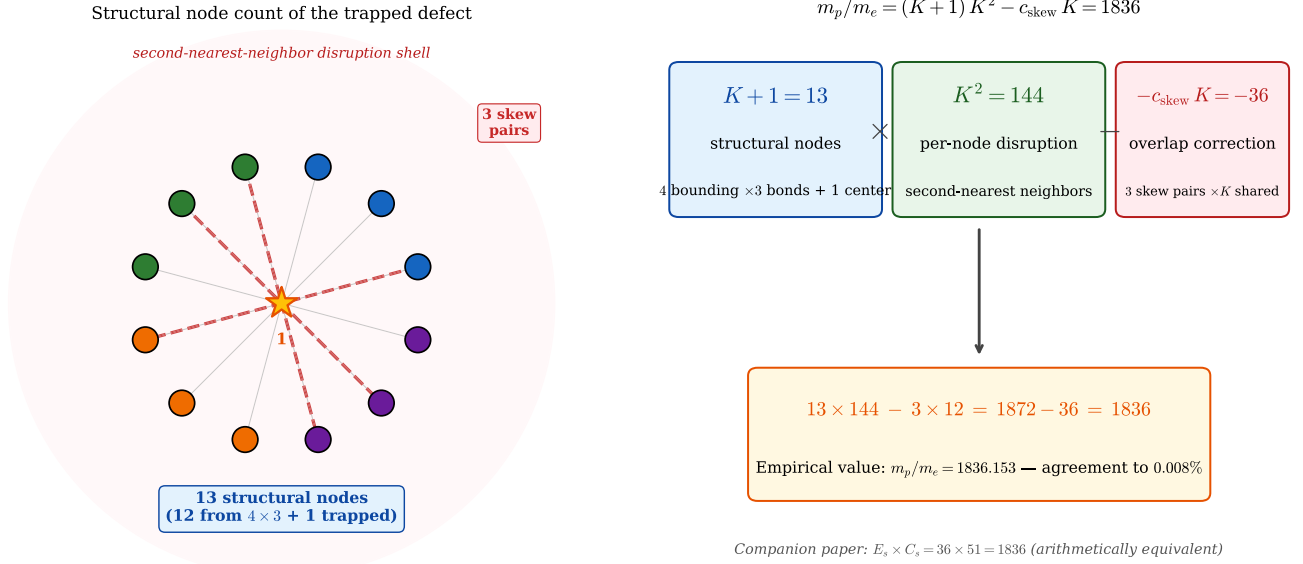


Fig. 5. The macroscopic topological disruption zone. Left: schematic of the 13 structural nodes of the localized defect — the central trapped node (yellow star) plus 12 surrounding bonded nodes (one per internal bond from each of the four bounding vertices, color-coded by parent vertex). The three skew-edge pairs are highlighted (red dashed). Right: the three-component derivation of m_p/m_e : 13 structural nodes (per Eq. (15)), each disrupting $K^2 = 144$ second-nearest-neighbor states, minus 36 double-counted bonds along the three skew-edge pairs, gives 1836 — matching empirical $m_p/m_e = 1836.153$ to 0.008%. The companion paper [2] reaches the same number through the dimension of the $[[192, 130, 3]]$ CSS-code coupling matrix (see Section 7.4).

8. The lighter particles

The proton-to-electron mass ratio derived in Section 7 uses the electron rest mass as the unit scale. Within the present framework, the electron is the minimal single-node defect of the $K=12$ lattice — a point defect localized at one bulk vertex, with no internal edges, no skew-edge pairs ($c_{\text{skew}} = 0$), and no fractional substructure — defining the minimal discrete strain unit of the vacuum. The other lighter particles (the pion, the muon, and the neutron) require fault-tolerant verification-cost arguments that lie outside the structural-disruption framework of this paper; they are treated in the companion paper [2], which derives $C_e = 1$, $C_\pi = 273$, $C_\mu = 207$, and $C_n = 1839$, all matching empirical mass ratios to within 0.12%.

9. Lorentz invariance and experimental constraints

A common concern about discrete-spacetime models is conflict with special relativity. We establish emergent Lorentz invariance in three steps and verify compatibility with current experimental bounds.

9.1. Spatial isotropy at the lattice level

The 12 nearest-neighbor bond vectors of the FCC lattice are

$$n_j \in \{(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)\} / \sqrt{2}, \quad j = 1, \dots, 12. \quad (19)$$

Define the rank-2 structure tensor

$$S_{\mu\nu} \equiv \sum_{j=1}^{12} n_j^\mu n_j^\nu. \quad (20)$$

Direct enumeration [2] gives $S_{xx} = S_{yy} = S_{zz} = 4$ and all off-diagonal components vanish. Therefore

$$S_{\mu\nu} = 4 \delta_{\mu\nu} \quad [\text{exact, by enumeration}]. \quad (21)$$

This guarantees equal propagation speed in every spatial direction.

The odd-rank tensor

$$T_{\mu\nu\lambda} \equiv \sum_{j=1}^{12} n_j^\mu n_j^\nu n_j^\lambda \quad (22)$$

vanishes exactly: every bond n_j has a partner $-n_j$ (the FCC lattice is centrosymmetric), so every odd-power sum cancels:

$$T_{\mu\nu\lambda} = 0 \quad [\text{exact, by inversion symmetry}]. \quad (23)$$

This ensures $\omega(k) = \omega(-k)$: there is no preferred direction and no linear (k) term in the dispersion.

9.2. Isotropy of the dispersion relation

For a scalar field on the FCC lattice, the dispersion is $\omega(k)^2 = \kappa \sum_{j=1}^{12} [1 - \cos(k \cdot n_j a)]$. At long wavelengths ($|k|a \ll 1$):

$$\omega(k)^2 \approx (\kappa a^2/2) \sum_{j=1}^{12} (k \cdot n_j)^2 = (\kappa a^2/2) k_\mu k_\nu S_{\mu\nu} = (\kappa a^2/2) \cdot 4 |k|^2 = 2\kappa a^2 |k|^2, \quad (24)$$

so $\omega = c_{\text{lat}} |k|$ with $c_{\text{lat}} = a\sqrt{2\kappa}$. The dispersion is exactly isotropic at all orders in k for which the Taylor expansion is valid. Corrections appear only at $O(k^4 a^4)$ from higher cosine terms, suppressed by $(|k|a)^4 \sim (E/M_P)^4$ for Planck-scale lattice spacing $a \sim \ell_P$.

9.3. Lorentz boosts in the continuum limit

The two tensor identities (Eq. (21), (23)) and the isotropic linear dispersion are sufficient for $\text{SO}(3,1)$ Lorentz boosts to emerge in the standard continuum limit [14]. A lattice Hamiltonian whose dispersion is isotropic and linear in k at long wavelengths gives rise to a relativistic effective field theory in the continuum limit. The lattice does break Lorentz symmetry at the cutoff scale $1/a$, but these violations are suppressed by $(E/M_P)^4$, far below current experimental sensitivity.

9.4. Experimental constraints

Current bounds on Lorentz violation are summarized in the Kostelecký–Russell data tables [15] and the Liberati–Maccione review [16]. Typical constraints fall in the range 10^{-20} to 10^{-40} relative to the Planck scale, depending on the operator and the experimental

probe. Our predicted suppression of $(E/M_p)^4 \sim 10^{-56}$ at optical frequencies is many orders of magnitude below these bounds, so the SSM is compatible with all current Lorentz-invariance tests.

9.5. Resolution of the Collins et al. naturalness objection

Collins et al. [14] pointed out that any Lorentz-violating UV cutoff generates fine-tuning of order Λ_{UV}^2/M_p^2 via radiative corrections, which would normally dominate over the bare suppression. In the SSM, this objection is addressed by the holographic structure of the lattice: the UV cutoff is effectively the 2D boundary network, which carries continuous $SO(2)$ symmetry. Radiative corrections in the boundary respect $SO(2)$ at every loop order, and the 3D bulk inherits this symmetry holographically. The full argument is given in [2], Section 2.2.

10. Discussion

10.1. What this picture gets right

From a single geometric premise — an extra node trapped in an interstitial tetrahedral void — we structurally obtain:

1. Fractional charges $-1/3$ and $+2/3$ from the regular-tetrahedron bond-angle cosine combined with integer winding under the bulk FCC Bravais translation symmetry. The full first-shell baryon spectrum (Δ^- , neutron, proton, Δ^{++}) follows from the four possible total winding numbers $W \in \{0, 1, 2, 3\}$ via $Q_{\text{baryon}} = -1 + W$.
2. Exactly three colors generated by the three independent skew-edge pairs.
3. Linear color confinement dictated by the metric wall at $L/\sqrt{3}$, reproducing the σr term of the Cornell potential.
4. $m_p/m_e = 1836$ derived from the structural node count and skew-edge pair count, with arithmetic equivalence to the $[[192, 130, 3]]$ CSS-code derivation in the companion paper [2].
5. The Δ^- , neutron, proton, and Δ^{++} spectrum from total winding number, and matter–antimatter duality from spatial inversion of the tetrahedral void.
6. Physical string breaking from lattice un-stitching limits.
7. Electromagnetic invisibility of quarks from the non-bipartite triangular face topology.
8. Compatibility with all current Lorentz-invariance experimental bounds.

None of these results require fine-tuned free parameters.

10.2. What remains open

Several aspects of the framework are not yet resolved:

- The complete meson spectrum beyond the pion.
- The geometric origin of the strange and heavier quark generations.
- The relationship between this static defect picture and the dynamical QCD flux tubes modeled by current gauge theories. The static topological structure provides the boundary conditions and the Coulomb-like short-distance term is expected to emerge from gauge dynamics, but a full derivation is not provided here.
- The proton–neutron mass splitting at the level of QFT corrections beyond the integer baseline.
- The phenomenological restriction of valence-bond windings to $w \in \{0, +1\}$ rather than the full integer ladder. The framework predicts the structure of the charge ladder rigorously but takes the truncation to lowest two states as input from the observed first-shell quark spectrum (Section 4.3).

10.3. Relation to other approaches

Lattice gauge theory. Lattice QCD [1,7] computes hadronic properties numerically on a background lattice using the Wilson action and its descendants. The SSM proposes that the lattice is not merely a mathematical computational tool but the physical vacuum itself. Confinement is therefore a static geometric property of the space rather than a dynamical property of a continuous gauge field living on the space. Our discrete-color and confinement arguments build on the standard lattice formulation reviewed by Kogut [8] and verified numerically by Creutz [17]. The treatment of fermions on the lattice — the Karsten–Smit no-go theorem on chiral invariance and species doubling [18] and the Susskind staggered-fermion construction [19] — defines the technical landscape within which a discrete-spacetime model must reproduce continuum physics. The SSM’s static topological derivation is complementary to (not a replacement for) the dynamical Wilson-loop approach, and the two should agree in the continuum limit.

Emergent spacetime and discrete geometry. The view that geometry itself emerges from a more primitive substrate has multiple complementary realizations in the recent literature. The ER=EPR conjecture of Maldacena and Susskind [20] identifies geometric connections (Einstein–Rosen bridges) with quantum entanglement; Van Raamsdonk [21] argues that classical spacetime connectivity is built up by entanglement; Swingle [22] maps entanglement renormalization onto holographic geometry; and the holographic quantum error-correcting codes of Pastawski et al. [23] make the emergence of bulk locality from boundary entanglement explicit. Other foundational discrete-spacetime programs include causal dynamical triangulations [24], causal sets [25], the spin-network combinatorics of Penrose [26], and the recent spacetime-quasicrystal construction of Boyle and Mygdalas [27]. The SSM differs from these in that the discrete substrate is a specific physical lattice (FCC) rather than a generic combinatorial graph, and its predictions are quantitative integer ratios of observable particle properties rather than asymptotic geometric features.

Regge calculus and discrete gravity. Regge’s foundational coordinate-free formulation of general relativity [4] provides the geometric machinery (the deficit angle around an edge as the discrete curvature) that we use to characterize the geometric frustration of the $K=4$ tetrahedral foam. Cheeger, Müller, and Schrader [28] placed Regge calculus on a rigorous footing by analyzing the curvature of piecewise flat spaces, and Hamber [29] provides a comprehensive treatment of lattice approaches to quantum gravity that share with the SSM the premise that spacetime admits a discrete description. The SSM uses these ideas in a constrained way: only the deficit-angle calculation enters, as the geometric quantity that drives the $K=4 \rightarrow K=12$ phase transition.

Topological quantum error correction. The interpretation of the FCC vacuum as a CSS code in our companion paper [2,12] is part of a broader research program that began with Kitaev’s anyon model for fault-tolerant quantum computation [30] and the topological-quantum-memory analysis of Dennis et al. [31]. The defining feature of these constructions is that information is protected not by an active measurement-and-correct cycle but by the topological properties of the underlying lattice. The SSM extends this perspective to the physical vacuum: the trapped tetrahedral defect is a topologically stable configuration whose inability to relax (the metric wall of Section 6) is the analog of the topological protection that makes anyonic codes useful.

Crystallization mechanisms. The framework’s central image — matter as incomplete crystallization of the vacuum — has structural analogs in two well-studied physical processes. Witten and Sander’s diffusion-limited aggregation [32] demonstrates that local kinematic rules generate persistent macroscopic non-uniformities that survive long after the rules cease to be locally informative. Coleman’s analysis of the false

Table 4

Kinematic parameters of the SSM lattice simulation. All values are geometrically or thermodynamically determined.

Parameter	Value	Derivation
Unitary metric (L)	1.0	Invariant relational distance
Lateral height	$\sqrt{3}/2 \cdot L \approx 0.866 L$	Equilateral triangle altitude
Lift height	$\sqrt{2/3} L \approx 0.816 L$	Regular tetrahedron altitude
Lift probability	$e^{-3} \approx 4.98\%$	Topological tunneling
Proximity bond	1.05 L	Regge deficit ($\delta \approx 7.36^\circ$)
Hard shell (R_{ex})	0.95 L	Symmetric exclusion
Geometric cutoff	$L/\sqrt{3} \approx 0.577 L$	Circumradius of unitary triangle

vacuum [33] establishes that a quantum field can become trapped in a metastable phase whose decay proceeds through nucleation events governed by topological barriers. The SSM combines these two ideas: the cosmological vacuum begins in a metastable $K=4$ tetrahedral phase, transitions to the $K=12$ FCC ground state through a Kepler-bound saturation process, and leaves behind the trapped extra nodes of an incomplete crystallization as the topological defects we identify with baryons.

11. Conclusions

Matter is spacetime that did not finish crystallizing. A single extra node trapped in a tetrahedral void of the $K=12$ FCC vacuum lattice strictly recovers fractional quark charges, three color degrees of freedom, linear confinement, and the proton-to-electron mass ratio — all emerging organically from the unadjusted geometry of close-packed spheres. The framework makes clear, falsifiable predictions: no isolated quarks can propagate, no fourth color charge can exist, and the first-shell baryon spectrum is exhausted by the four configurations $W \in \{0, 1, 2, 3\}$ corresponding to Δ^- , neutron, proton, and Δ^{++} . The model is consistent with all current experimental bounds on Lorentz violation and reproduces the linear-confinement piece of the Cornell potential. The companion paper [2] reaches $m_p/m_e = 1836$ by a different route (fault-tolerant verification cost in a CSS code on the same lattice); the two derivations are arithmetically equivalent because $K = 4c_{\text{skew}}$ in the cuboctahedral geometry, and the structural meaning of that equivalence is left for future work.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix. Kinematic parameters and replication conditions

This appendix specifies the kinematic parameters of the simulation reported in Section 2.3 and the conditions under which its quantitative claims (Table 2 and Fig. 2) are reproduced. The kinematic operators (stitch and lift) and their geometric origins are defined in Section 2.3; this appendix collects the operational parameters in a single table for ease of reference.

A.1. Kinematic parameters

See Table 4.

A.2. Replication

The kinematic rules described in Section 2.3 (Bell-pair seed; stitch and lift operators with relative amplitude $P_{\text{lift}} = e^{-3}$; proximity bonding within 1.05 L ; exclusion radius $R_{\text{ex}} = 0.95 L$), together with the parameter values in Table 4, fully specify the simulation. Running the

resulting graph-growth procedure at $N = 1000$ over 30 random seeds reproduces the data in Table 2: $K=12$ saturation of $25.4 \pm 5.4\%$, mean coordination $\bar{K} = 8.90 \pm 0.26$, and shape isotropy $\lambda_{\text{min}}/\lambda_{\text{max}} = 0.54 \pm 0.15$. The finite-size scaling exponent is $\alpha = 6.8 \pm 0.6$ (Eq. (2)). A sweep over $R_{\text{ex}} \in [0.50, 1.02]$ at fixed $N = 500$, $P_{\text{lift}} = 0.05$ reproduces Fig. 2: the sharp phase transition at $R_{\text{ex}} = L/\sqrt{3}$ and the wide stability plateau in $[0.58, 0.99]$. All quantitative claims in Section 2.3 reduce to these two scans.

A reference implementation of the algorithm is available at `ssm_sim.py` (lattice generation and saturation analysis); the exclusion-radius sweep is performed by `plot_rex_sweep.py`. The implementation is provided as a verification aid; the manuscript itself fully specifies the algorithm and any equivalent implementation will reproduce the results within the reported statistical uncertainties.

Data availability

All kinematic parameters and replication conditions for the simulation results in Section 2.3 are given in Appendix. A reference implementation of the algorithm is available as `ssm_sim.py` (the lattice generator and saturation analysis used for Table 2) and `plot_rex_sweep.py` (the exclusion-radius sweep used for Fig. 2). The bulk-interior diagnostic reported in Table 2 is computed by `ssm_bulk_analysis.py`, which re-analyzes the simulation output to extract bulk-restricted co-ordination statistics. The interactive 3D visualizations are available at https://raghu91302.github.io/ssmtheory/ssm_regge_deficit.html and https://raghu91302.github.io/ssmtheory/ssm_quark_structure.html.

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